

Aim & Objective



Aim

Design a compressor-based water cooler dispenser that efficiently delivers clean, safe, and continuously cooled drinking water.

Objective

Apply vapor compression refrigeration for water cooling.

Build a working prototype with suitable materials.

Evaluate cooling efficiency, energy use & temperature stability.

Conclusion

Successfully built a compressor-based water cooler dispenser.

Delivers clean, cool water using vapor compression refrigeration.

Safe, energy-efficient, and easy to operate.

Future improvements: solar power, auto temperature control, enhanced insulation.



Auto temperature control Water Cooler



Solar powered Water cooler



TECHNOLOGICAL UNIVERSITY
(MANDALAY)



DEPARTMENT OF MECHANICAL
ENGINEERING

“Water Cooler Dispenser”



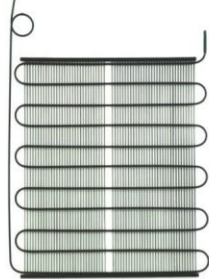
Components of Water Cooler Dispenser And Their Functions

Compressor



A compressor is a device that operates mechanically to increase the pressure of a gas through volume reduction.

Condenser



The condenser is a vital component in a compressor-based water cooler dispenser, responsible for releasing heat absorbed by the refrigerant during the cooling process.

Capillary Tube



The capillary tube is one of the essential expansion devices that are found in small refrigeration systems, for instance, water cooler dispensers that are based on compressors.

Evaporator Coil



The evaporator coil is one of the main components of a water cooler dispenser. Its main function is to absorb heat from the water and make the water cold.

Working Procedure of Water Cooler Dispenser

- Water enters the storage tank.
- The compressor compresses the refrigerant and sends it to the condenser.
- The condenser removes heat from the refrigerant.
- The refrigerant passes through the capillary tube, reducing its pressure and temperature.
- The cold refrigerant flows through the evaporator coil and absorbs heat from the water.
- The cooled water is stored in the tank and dispensed through the outlet when needed.

Components Assembly and Frame Setting

